Feedback-Induced Distribution Shift in Multilingual LLM Deployment: Evidence, Characterization, and RL-Based Adaptation Strategies

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Abstract

Large language models deployed across multilingual production environments often experience performance degradation for underrepresented languages over time. I identify this phenomenon as Feedback-Induced Distribution Shift (FIDS), a systems-level failure in which design and evaluation choices allow localized quality declines to persist while aggregate metrics remain stable.

Drawing on production-scale experience in multilingual LLM deployment, I characterize three recurring patterns through which these failures emerge, give each pattern a formal proof-backed guarantee, and show that standard benchmarking approaches are poorly suited to detect them.

To address this, I introduce RL-Stable Deployment (RLSD), a reinforcement learning-based framework that dynamically adapts inference policies in response to performance signals, without requiring model retraining.

I argue that these failures arise primarily from system design rather than model limitations, and that addressing them requires new evaluation and deployment paradigms beyond improved benchmarks.

1. Introduction

Over four years of managing multilingual AI systems at scale, I repeatedly observed a consistent pattern: features that performed well in primary launch markets degraded significantly in underrepresented locales. While aggregate quality metrics indicated acceptable performance, failures in Tier 2 and Tier 3 markets often went undetected. These failures were not subtle: queries failed to trigger features, and outputs were clearly incorrect to native speakers, yet they remained invisible in aggregated evaluation frameworks.

The consequences were cumulative. Users in these markets engaged less with the system, resulting in reduced feedback signals. This, in turn, limited opportunities for system improvement, reinforcing and amplifying existing quality gaps over time.

Importantly, this pattern did not arise from intentional neglect. Rather, it reflects structural properties of how systems are designed and evaluated: metrics are optimized for primary markets, engineering effort follows measurable impact, and low-signal populations remain underrepresented in both evaluation and prioritization pipelines. The feedback loop exists, but it systematically disadvantages the very users who experience the poorest performance.

To formalize this phenomenon, I introduce Feedback-Induced Distribution Shift (FIDS). In contrast to traditional notions of distribution shift driven by changes in data or model updates, FIDS arises from the interaction between deployed models and the feedback systems surrounding them, specifically, evaluation

metrics, prioritization processes, and organizational incentives. The distribution shifts not because the model changes, but because the system selectively amplifies some user signals while suppressing others.

FIDS is related to, but distinct from, prior work on data feedback loops (e.g., Taori and Hashimoto, 2023), which focuses on how model outputs influence future training data. In contrast, FIDS operates at the level of deployment and system design, and can emerge even in the absence of model retraining. It also sits within the broader landscape of foundation model risk surveyed by Bommasani et al. (2021), though FIDS isolates a specific deployment-time mechanism rather than cataloguing risk categories.

This paper makes four contributions. First, I present a taxonomy of FIDS consisting of three subtypes, each grounded in recurring operational patterns. Second, for each subtype I give a minimal formal model and prove a corresponding proposition, establishing, rather than merely illustrating, why the failure is structural and not incidental. Third, I analyze the conditions under which these subtypes become most severe and least detectable. Fourth, I introduce RL-Stable Deployment (RLSD), an inference-time framework for identifying and mitigating FIDS without requiring model retraining.

Rather than presenting a fully closed solution, this work aims to formalize a class of failures that are widely observed in practice but insufficiently captured by existing evaluation paradigms.

2. Background and Related Work

2.1 Tiered Deployment and Its Quality Consequences

Multilingual AI systems are commonly deployed using a tiered market strategy, in which initial releases prioritize a small set of high-engagement markets (e.g., the United States, United Kingdom, and other economically dominant regions). These markets typically receive the most comprehensive feature coverage, engineering investment, and pre-launch quality validation. Subsequent deployments extend to lower-tier markets with reduced validation and, in many cases, adaptations of existing features rather than fully localized implementations. While this approach reflects a practical allocation of limited resources, it introduces systematic disparities in product quality that are often insufficiently measured or tracked over time.

A key consequence of tiered deployment is that features deployed in lower-tier markets are frequently calibrated to the query distributions and usage patterns of higher-tier markets. As a result, they may fail to generalize to local linguistic structures, cultural contexts, or region-specific user intents. For example, a feature that reliably triggers for English queries with relatively fixed syntactic patterns may exhibit degraded or inconsistent behavior when exposed to morphologically richer or less standardized language forms.

These failures are not necessarily indicative of deficiencies in the underlying model. Rather, they reflect a mismatch between the conditions under which the system was developed and the environments in which it is deployed. In this sense, tiered deployment introduces a form of deployment-induced error, where system performance degrades due to insufficient adaptation to local contexts rather than inherent model limitations.

2.2 Distributionally Robust Machine Learning

Prior work in distributionally robust machine learning has highlighted the limitations of optimizing for average-case performance in heterogeneous data settings. Sagawa et al. (2020) show that models trained using standard empirical risk minimization can achieve strong aggregate performance while exhibiting significantly worse outcomes on underrepresented subgroups. Similarly, Hashimoto et al. (2018) propose a distributionally robust optimization (DRO) framework that improves worst-case subgroup performance without requiring explicit group annotations, and Oren et al. (2019) extend this DRO framing specifically to language modeling, showing that standard training objectives can leave worst-case topic or domain performance unbounded even as average perplexity improves.

This line of work is directly relevant to the tiered deployment dynamics described in Section 2.1, as both highlight systematic disparities that are obscured by aggregate evaluation metrics. However, the focus of these approaches is primarily on mitigating bias during training by adjusting optimization objectives or data weighting strategies.

In contrast, the phenomenon described in this paper, Feedback-Induced Distribution Shift (FIDS), emerges after deployment. It arises not from the training distribution itself, but from the interaction between a deployed model and the feedback systems that govern its evaluation and iteration. As such, FIDS reflects a form of deployment-time bias that compounds through measurement practices, prioritization decisions, and uneven feedback signals, rather than through the model's learned parameters alone.

2.3 LLM Evaluation Frameworks

Liang et al. (2022) introduce HELM (Holistic Evaluation of Language Models), a comprehensive framework for evaluating large language models across a range of scenarios, metrics, and user populations. HELM's multilingual evaluations highlight systematic performance differences across languages and task settings, providing an important baseline for understanding cross-lingual variability in model behavior.

However, HELM evaluates model performance at a fixed point in time under controlled and standardized conditions. As a result, it does not capture how performance evolves after deployment, particularly in production environments where evaluation, prioritization, and iteration are shaped by real-world usage patterns and organizational constraints.

The phenomenon described in this paper builds directly on these observations. While HELM surfaces cross-lingual performance disparities at evaluation time, Feedback-Induced Distribution Shift (FIDS) characterizes how such disparities can persist, amplify, or become less visible over the operational lifetime of a system. In this sense, static evaluation frameworks provide an initial snapshot of quality differences, whereas FIDS focuses on their dynamic evolution under deployment conditions.

2.4 Organizational Dynamics in AI Development

The influence of organizational incentives on AI system quality remains relatively underexplored in the technical literature, despite its practical importance in large-scale deployments. Prior work on machine learning systems has examined forms of technical debt that arise from increasing system complexity and maintenance burdens. For example, Sculley et al. (2015) and Henderson et al. (2018) highlight how

design and engineering decisions can introduce long-term costs that are not immediately visible at deployment time.

However, these frameworks primarily focus on code, data pipelines, and model behavior, and do not explicitly address how organizational priorities and incentive structures shape the distribution of system quality across different user populations. In practice, factors such as feature velocity, market prioritization, and competitive pressure can influence which issues are addressed and which are deferred, potentially leading to uneven quality outcomes across regions and languages.

The framework introduced in this paper builds on this gap by examining how such dynamics affect multilingual deployments in particular. While prior work has not, to my knowledge, formally characterized these effects in this context, the Aspirational Deferral Lock (ADL) subtype proposed later in this paper provides one mechanism, stated as a proposition with a short proof in Section 3.4, through which organizational incentives can contribute to persistent and compounding quality disparities.

3. A Taxonomy of Feedback-Induced Distribution Shift

Feedback-Induced Distribution Shift (FIDS) can be decomposed into three recurring subtypes, each corresponding to a distinct underlying mechanism and observable pattern in deployed systems. These subtypes differ in how quality degradation emerges, how it manifests in production telemetry, and how it must be detected and mitigated in practice.

While analytically separable, the subtypes are not independent. In real-world deployments, they often co-occur and interact, leading to compounding effects that amplify disparities across user populations. This interaction is a key feature of FIDS: localized failures are not isolated, but can reinforce one another through shared feedback and prioritization dynamics.

The taxonomy presented here is derived from recurring patterns observed across multilingual deployments. Below, each subtype is given both an operational description and a minimal formal model with a proven proposition, so that the claim “this failure is structural” is a mathematical statement rather than an inference from anecdote.

3.1 Setup and Notation

Let there be L locales, indexed i = 1, …, L. At time t, let n_i(t) denote query volume (traffic) from locale i, and let q_i(t) ∈ [0,1] denote the quality of the deployed system for locale i (accuracy, triggering precision, or any bounded quality metric). Let N(t) = Σ_i n_i(t) be total traffic. The aggregate metric reported on dashboards is the traffic-weighted average of per-locale quality:

```
Q_agg(t) = (1 / N(t)) · Σ_{i=1}^{L} n_i(t) q_i(t)
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This is not a simplification of convenience; in my experience across production multilingual systems, it is commonly how “topline” quality metrics are computed, because raw per-query accuracy is what is measured, and it gets rolled up by summing over queries. The three propositions below all take Q_agg, or a component of the system that produces it, as their object of study.

3.2 Aggregate Masking Effect (AME)

The Aggregate Masking Effect (AME) arises when system performance is monitored using aggregate metrics that obscure disparities across user populations. Because these metrics are typically weighted by traffic volume, they disproportionately reflect the performance experienced by majority groups, while masking degradation in lower-volume populations.

The mechanism follows directly from the structure of weighted aggregation. Consider a deployment in which a dominant market accounts for the majority of traffic, while smaller markets contribute a relatively small fraction. An aggregate metric, such as accuracy, engagement, or user satisfaction, will be dominated by the performance of the highest-weight group. As a result, substantial quality gaps in lower-weight groups may have only a negligible effect on the reported metric. For example, a system that achieves 94% accuracy for a high-traffic group and 61% accuracy for a lower-traffic group may still report aggregate accuracy above 90%, depending on the traffic distribution.

Proposition 1 (Aggregate Masking Effect).

An arbitrarily severe quality collapse in a subset of locales can produce an arbitrarily small change in the reported aggregate metric, provided that subset's traffic share is small enough.

Setup. Suppose a subset S of locales degrades: $q_i \to q_i - \delta_i$ for $i \in S$, with $\delta_i \le \delta_{max}$, while all other locales are unchanged. Let $w_S = (1/N) \Sigma_{i \in S} n_i$ be the traffic share of the degraded locales.

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Q_agg^before − Q_agg^after = (1 / N) · Σ_{i∈S} n_i δ_i ≤ δ_max · w_S
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Proof. Direct substitution: only terms for $i \in S$ change, each by at most $n_i \cdot \delta_{max}$, so the total change is at most $\delta_{max} \cdot \Sigma_{i \in S} n_i = \delta_{max} \cdot N \cdot w_S$; dividing by N gives the bound. ■

Why this matters. If a monitoring system alerts only when Q_{agg} drops below threshold τ, Proposition 1 shows the alert cannot fire, regardless of how severe the underlying degradation is (δ_i can be as large as 1), whenever $w_S < \tau/\delta_{max}$. Masking is not a monitoring bug to be tuned away; it is a direct, provable consequence of using a traffic-weighted average as the sole detection signal. In practice, this results in a failure mode where system-level metrics indicate acceptable performance while meaningful segments of the user population experience degraded or inconsistent behavior: reduced feature activation rates, misalignment with local linguistic patterns, or degraded response quality, none of which are captured by aggregate reporting alone.

Mitigation strategies, such as disaggregated evaluation and locale-specific alerting, are well understood in principle. However, in practice, they are often underutilized, in part because surfacing fine-grained performance gaps introduces additional prioritization and engineering overhead, a dynamic formalized separately as Aspirational Deferral Lock in Section 3.4.

3.3 Low-Resource Locale Collapse (LRLC)

Low-Resource Locale Collapse (LRLC) describes a self-reinforcing degradation process that can emerge once performance disparities are no longer visible in aggregate monitoring. Building on the masking effect described in Section 3.2, LRLC captures how unaddressed quality gaps in lower-resource locales can compound over time through feedback-driven dynamics.

The core mechanism is a negative feedback loop between system quality and user engagement. When performance in a given locale degrades, users are less likely to engage with the affected features. Reduced engagement leads to diminished data signals, such as fewer interactions, lower-quality feedback, or reduced coverage of relevant query patterns, which in turn limits opportunities for system improvement. This reduced improvement capacity allows performance gaps to persist or widen, further suppressing engagement.

This process produces a characteristic signature in production data. Over time, query volume and feature usage in affected locales may decline, either because users disengage entirely or adapt their behavior to avoid unreliable functionality. As a result, the system receives progressively less informative signal from these populations. In many cases, by the time degradation becomes visible in aggregate metrics, the affected user base has already contracted substantially.

Proposition 2 (Low-Resource Locale Collapse).

Locales with less human-curated training data degrade under feedback in a way that well-resourced locales do not, even when both start from similar quality.

Setup. Let training data for locale i at time t+1 be a mixture of human-verified examples (quality q_human, fixed) and model-generated examples, with mixture weight $\alpha_i(t) \in [0,1]$ on the model-generated share. In practice α_i rises as human annotation capacity for locale i falls; this is the discrete-time analogue of the continuous decline in q(t) described operationally above. Following the model-collapse literature (Shumailov et al., 2023), assume model-generated examples have expected quality $q_i(t) - \varepsilon$ for some $\varepsilon \geq 0$ reflecting compounding self-reference. Then:

```
q_i(t+1) = (1 − α_i) q_human + α_i (q_i(t) − ε)
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Result. This is a linear recursion in $q_i(t)$ with fixed point $q_i^* = q_{human} - [\alpha_i/(1-\alpha_i)]\varepsilon$ when $\alpha_i < 1$, approached geometrically at rate α_i. As $\alpha_i \to 1$ (locale almost entirely dependent on model-generated data), the fixed point drops without bound in ε; the well-resourced case ($\alpha_i \to 0$) stays anchored near q_{human} regardless of ε.

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q_i* = q_human − [α_i / (1 − α_i)] ε
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Proof. Standard linear recursion analysis: $q_i(t+1) = \alpha_i q_i(t) + [(1-\alpha_i)q_{human} - \alpha_i\varepsilon]$ has the fixed point above, and converges to it since $0 \leq \alpha_i < 1$. ■

Why this matters. This gives LRLC a precise mechanism: it is not just “less data, worse model”; it is that the composition of what little data exists shifts toward self-generated content specifically where human annotation investment is weakest, and that shift is what compounds. A defining feature of LRLC is that its early stages are often obscured by the Aggregate Masking Effect (Proposition 1): a locale's contribution to Q_{agg} stays small even as α_i rises, so intervention is delayed until the feedback loop has significantly reduced both q_i and the human-verified share available to correct it, making recovery more difficult. This is also the direct theoretical bridge to the RL-based adaptation strategies of Section 4: a policy that raises the human-verified share $(1-\alpha_i)$ specifically where α_i is high is provably sufficient to keep q_i^* bounded away from collapse.

3.4 Aspirational Deferral Lock (ADL)

Aspirational Deferral Lock (ADL) describes the organizational dynamics that prevent the remediation of persistent quality gaps, even after they have been identified. While Aggregate Masking Effect (AME) obscures disparities and Low-Resource Locale Collapse (LRLC) drives their compounding, ADL explains why these issues often remain unresolved in practice.

ADL arises when quality improvement efforts for underrepresented user populations are consistently deprioritized relative to the development of new features. In many large-scale AI deployments, identified issues are placed into backlogs or long-term planning categories without clear ownership, timelines, or allocated resources. Over time, this leads to the accumulation of what can be described as quality debt, unresolved performance deficiencies that disproportionately affect specific user groups.

A distinguishing aspect of ADL is the absence of a concrete resolution pathway. Unlike traditional notions of technical debt, which typically imply eventual remediation, deferred quality issues in this context may persist indefinitely due to shifting priorities and limited engineering capacity. This dynamic is often reinforced in environments where success is measured primarily by feature velocity, product expansion, or competitive positioning, rather than by uniformity of user experience across populations.

Proposition 3 (Aspirational Deferral Lock).

A backlog item can have organization-wide consensus that it is real and severe, and still have zero probability of ever being fixed under a greedy resource-allocation rule, if it carries zero weight in the objective function driving that allocation.

Setup. Model engineering-effort allocation as a greedy scheduler: at each step, one unit of effort goes to the locale maximizing $w_i \cdot g_i$, where $g_i \geq 0$ is the marginal quality gain from fixing locale i's backlog item (the “everyone agrees it's real” term), and $w_i \geq 0$ is that locale's weight in the organization's objective, in practice, a function of which team's OKRs or traffic tier the locale falls under.

```
arg max_i ( w_i · g_i )
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Result. If $w_i = 0$ for locale i, then locale i is never selected by this rule, regardless of g_i, so long as some other locale i' has $w_{i'} > 0$.

Proof. For any i with $w_i = 0$: $w_i \cdot g_i = 0 \leq w_{i'} \cdot g_{i'}$ for any i' with $w_{i'} > 0$ and $g_{i'} \geq 0$, so i is dominated in every round. ■

Why this matters. This is deliberately a simple statement, because the phenomenon is simple: the “aspirational backlog” is not a prioritization failure in the sense of bad judgment; it is what greedy allocation under a zero-weight objective always produces, as a matter of the algorithm's structure, independent of how large g_i is. Formally, this recovers the informal backlog dynamic $B(t)$ with discovery rate r and resolution rate f: whenever the objective assigns $w_i = 0$ to a locale, that locale's f is identically zero regardless of r, so $B(t)$ for that locale grows without bound. The implied fix is direct: ADL only breaks if w_i is made a function of backlog age or locale coverage rather than current traffic, exactly the kind of policy-level RL adaptation strategy explored in Section 4.

The interaction between ADL and LRLC is particularly consequential. As performance degrades in a given locale and engagement declines (Proposition 2), the resulting reduction in user activity further lowers n_i and hence w_i in most real objective functions, which by Proposition 3 further reduces the

probability that locale is ever selected for remediation. This creates a self-reinforcing cycle in which the populations experiencing the lowest quality also generate the weakest signals for intervention.

3.5 Connection to Section 4

The three propositions above are exactly what the RLSD framework in Section 4 is designed to act against: Proposition 1 predicts when a monitoring threshold fails to fire; Proposition 2 predicts the quality trajectory under different α_i regimes and the fixed point it converges to; Proposition 3 predicts which backlog items a greedy scheduler will starve. RLSD's three interventions in Section 4.2 map directly onto these three failure modes.

4. RL-Stable Deployment (RLSD): An Inference-Time Adaptation Framework

4.1 Why Inference-Time Adaptation

The failure modes described in Section 3 arise from a combination of technical and organizational factors. In principle, they can be addressed through improved monitoring practices, targeted investment in underrepresented locales, and organizational prioritization of quality alongside feature development. However, such interventions often require sustained coordination, resource allocation, and changes in incentive structures, which may be difficult to implement in practice, particularly in fast-moving or resource-constrained environments.

RL-Stable Deployment (RLSD) is designed as a complementary, system-level mitigation that operates within these constraints. Rather than relying on model retraining or organizational restructuring, RLSD introduces an inference-time adaptation layer that adjusts model behavior in response to signals derived from production telemetry.

At a high level, RLSD uses reinforcement learning to incorporate feedback signals, such as discrepancies in engagement, feature activation rates, or performance proxies across user populations, into the inference process. These signals are used to guide adaptive policies that modify how the model responds under different conditions, with the goal of reducing disparities that would otherwise persist undetected or unaddressed. This use of a learned feedback signal to shape inference-time behavior follows the same reinforcement-learning-from-feedback logic established for alignment more generally (Christiano et al., 2017), applied here to cross-locale performance parity rather than preference alignment.

Importantly, RLSD does not eliminate the underlying causes of FIDS. Instead, it provides a mechanism for mitigating their downstream effects in deployed systems, particularly in settings where more direct interventions are not immediately feasible.

4.2 Framework Design

RL-Stable Deployment (RLSD) introduces a lightweight control layer that operates alongside a deployed language model and continuously ingests production telemetry. This telemetry includes locale-disaggregated query distributions, concretely, per-locale $n_i(t)$ and estimates of $q_i(t)$ as defined in Section 3.1, engagement rates, feature activation statistics, and available proxies for user satisfaction. Rather than relying on the aggregate Q_{agg}, RLSD maintains a structured view of system performance across user populations over a rolling time window.

The RLSD controller can be formulated as a contextual decision process. At each timestep, the controller observes a state s_t composed of telemetry-derived signals (e.g., per-locale engagement trends, trigger rates, and relative performance disparities). Based on this state, it selects an action a_t from a constrained set of inference-time interventions. The objective is to optimize a reward signal that balances overall system performance with penalties for cross-locale disparities, thereby encouraging more uniform quality across populations.

These interventions are designed to be lightweight and composable within existing inference pipelines. They include:

- **Prompt adaptation:** applying locale-specific prefixes or instruction tuning at inference time
- **Routing strategies:** directing queries to specialized model variants or calibrated pipelines
- **Threshold adjustments:** modifying feature triggering conditions based on observed locale behavior

The mapping between FIDS subtypes and interventions follows directly from the propositions in Section 3:

AME (Proposition 1): RLSD monitors disaggregated $q_i(t)$ rather than only Q_{agg}, and identifies cases where a locale's degradation δ_i is large even though w_S is too small for it to move Q_{agg} past threshold τ. When such discrepancies are detected, the controller prioritizes targeted interventions, localized prompt adjustments or routing, for the affected group without altering global behavior.

LRLC (Proposition 2): RLSD tracks the effective mixture weight α_i implied by observed engagement and annotation trends. Sustained increases in α_i, particularly when correlated with degraded q_i, are treated as indicators of an approaching fixed point q_i^* below an acceptable bound. In response, RLSD increases the effective representation of these locales within the inference pipeline and triggers auxiliary processes such as data augmentation or targeted evaluation, interventions that act directly to lower α_i, the quantity Proposition 2 identifies as the lever.

ADL (Proposition 3): Because ADL originates from organizational dynamics external to the model, RLSD cannot alter w_i directly. Instead it focuses on improving visibility: the system generates structured reports that quantify persistent quality gaps across locales, in effect, estimates of g_i for every locale with $w_i = 0$, including their estimated impact on engagement and retention. By translating quality disparities into operational metrics, these reports aim to reduce the likelihood that such issues remain indefinitely deprioritized.

Overall, RLSD does not replace existing training or evaluation workflows. Instead, it augments deployed systems with a feedback-sensitive control layer that enables localized adaptation and earlier detection of emerging disparities.

4.3 Theoretical Properties

RL-Stable Deployment (RLSD) is designed to provide stability guarantees for worst-case performance across user populations under the failure modes described in Section 3. Let $W(t) = \min_i q_i(t)$ denote the minimum performance achieved across all locale groups at time t.

In the absence of targeted intervention, the combined effects of AME (Proposition 1), LRLC (Proposition 2), and ADL (Proposition 3) can lead to sustained degradation in worst-case performance: AME allows a locale to degrade without tripping an aggregate alert, LRLC gives that degradation a self-reinforcing trajectory toward $q_i^* < q_{human}$, and ADL guarantees the responsible backlog item is never selected for remediation once $w_i = 0$. In particular, when feedback signals from low-resource locales are weak or systematically deprioritized, performance in these groups may decline over time and fail to recover without external correction.

RLSD mitigates this behavior by introducing periodic monitoring and adaptive intervention. Under the assumption that (i) degradation in any locale is detectable through telemetry signals with bounded delay, and (ii) the intervention set includes actions capable of partially correcting observed disparities, the system can maintain a lower bound on worst-case performance. Formally, for monitoring frequency f, RLSD ensures that

$$W(t) \geq W(0) - \delta(f)$$

where $\delta(f)$ is a function that captures the maximum allowable degradation between successive interventions. As the monitoring frequency increases, the time between corrective actions decreases, leading to a tighter bound on worst-case performance.

This guarantee is inherently conditional: it depends on the quality of telemetry signals, the responsiveness of the intervention mechanism, and the effectiveness of available actions. Unlike Propositions 1–3, it is stated as an operating principle rather than proved from a fully specified model; closing that gap in the style of the distribution-free guarantees developed for conformal factuality (Mohri & Hashimoto, 2024), deriving $\delta(f)$ explicitly as a function of the detection-delay and intervention-strength assumptions, is identified as future work in Section 6.3. Nevertheless, it provides a useful operational target: sufficiently frequent and targeted intervention can prevent unbounded degradation in underrepresented populations.

Importantly, the guarantee is defined over worst-case locale performance rather than aggregate metrics. This distinction is central to the FIDS framework: maintaining high average performance does not preclude severe degradation in specific user groups. By focusing on the minimum performance across populations, RLSD directly targets the failure modes that aggregate evaluation obscures.

5. Why Manual Workarounds Are Not Enough

Prior to the development of a formal framework, mitigation efforts for multilingual quality gaps often relied on manual, localized interventions. These included prompt-level adjustments, tuning of feature triggering thresholds, and calibration of precision–recall tradeoffs to better align with the requirements of specific markets. In many cases, such interventions were effective in improving performance for individual features and locales.

However, these approaches exhibit fundamental limitations. First, they are inherently reactive: they depend on the prior identification of a problem, which in turn requires access to disaggregated performance data that is not always available in standard monitoring pipelines, precisely the data Proposition 1 shows is necessary, since Q_{agg} alone cannot surface it. Second, they are localized and non-transferable; each intervention addresses a specific feature–locale combination without providing a

general mechanism for detecting or mitigating similar issues elsewhere. Third, they lack persistence. Without continuous maintenance, improvements degrade over time as systems evolve and underlying conditions shift.

These limitations make manual approaches difficult to scale in large, multilingual deployments. Even when effective in isolation, they do not provide a systematic way to monitor, prioritize, and sustain quality improvements across diverse user populations.

RL-Stable Deployment (RLSD) is designed to address this gap. Rather than replacing expert-driven interventions, RLSD provides the infrastructure required to make such interventions consistent and scalable. By enabling continuous detection of performance disparities and supporting targeted, inference-time adaptation, RLSD transforms what would otherwise be ad hoc remediation into a structured, system-level process.

6. Discussion

6.1 The Measurement Problem

A central implication of the FIDS framework, and of Proposition 1 in particular, is that the associated failure modes are often not captured by commonly used aggregate quality metrics. Standard evaluation practices typically emphasize average performance across users, which can obscure substantial disparities between user populations.

Addressing this issue is not simply a matter of introducing additional metrics into existing dashboards. Rather, it requires a shift in how multilingual AI quality is conceptualized. In particular, evaluation frameworks must account for performance variation across populations, rather than relying solely on aggregate summaries.

This shift has practical consequences. Consider a system that achieves 95% average accuracy while delivering substantially lower performance, e.g., 50% accuracy, for a minority subset of users. Aggregate reporting would characterize this system as high-performing, despite the fact that a nontrivial portion of its user base experiences significantly degraded outcomes.

From the perspective of multilingual system evaluation, such a characterization is incomplete. A more appropriate assessment must consider both overall performance and the distribution of that performance across user groups, including the worst-case outcomes $W(t)$ introduced in Section 4.3. Without this perspective, evaluation metrics risk overstating system quality and obscuring disparities that are central to the failures described by FIDS.

6.2 The Novelty Trap

Proposition 3 (Aspirational Deferral Lock) highlights the role of organizational incentive structures in shaping how AI system improvements are distributed across user populations. In many competitive development environments, success is primarily measured through the introduction of new capabilities, user growth, and market differentiation. As a result, engineering effort tends to be directed toward feature expansion rather than systematic improvement of existing functionality.

This dynamic has uneven consequences. Efforts to improve quality for underrepresented user groups, such as speakers of lower-resource languages or users in lower-tier markets, often compete with initiatives that have more immediate and visible impact on headline metrics. Because these populations may contribute less to aggregate engagement or revenue signals, issues affecting them are less likely to be prioritized, even when they are known to be real, exactly as Proposition 3 predicts for any locale with $w_i = 0$.

Importantly, this pattern does not depend on individual intent. It emerges from the alignment between organizational incentives and the metrics used to evaluate success. When these metrics do not adequately capture disparities in user experience, optimization processes can systematically overlook segments of the population that fall outside dominant usage patterns.

Within this context, FIDS can be understood as a consequence of optimizing a complex sociotechnical system under incomplete measurement. When key aspects of user experience are not represented in the objective function, whether explicitly or implicitly, feedback loops may reinforce existing disparities rather than reduce them.

6.3 Limitations and Future Work

This work has several limitations that point to important directions for future research.

First, the proposed FIDS taxonomy (AME, LRLC, ADL) is derived from observed patterns in large-scale deployments rather than from a controlled empirical study spanning multiple organizations or system architectures. Propositions 1–3 establish that each mechanism is structurally capable of producing the described failure under the stated assumptions; they do not establish how often, or how severely, each mechanism dominates in any particular deployment. The taxonomy should be viewed as a conceptual framework, formally grounded but requiring further empirical validation across diverse deployment contexts.

Second, RL-Stable Deployment (RLSD) is presented as a system design and has not yet been implemented or evaluated in a production environment. The bound in Section 4.3 is stated as an operating principle rather than derived from the same level of formal assumption as Propositions 1–3; deriving $\delta(f)$ explicitly, and characterizing the conditions under which the detectability and intervention-effectiveness assumptions hold, is left to future work.

Third, the proposed approach to addressing Aspirational Deferral Lock (ADL) emphasizes improved measurement and visibility through mechanisms such as locale-level quality reporting. Proposition 3 shows that a nonzero w_i is necessary for a locale to ever be selected under greedy allocation; it does not show that visibility alone is sufficient to make an organization set $w_i > 0$. The relationship between measurement and organizational action is complex, and increased visibility alone may not be sufficient to drive sustained remediation.

Future work should address these limitations along several dimensions. Empirically, the FIDS framework can be evaluated by analyzing multilingual system performance across multiple deployment settings to assess the prevalence and interaction of the proposed subtypes. From a systems perspective, implementing RLSD and measuring its impact on both aggregate and worst-case performance would provide critical validation of its practical utility. Finally, further research is needed to understand how

organizational structures and incentive mechanisms can be aligned with quality objectives in multilingual systems.

7. Conclusion

This paper is motivated by a recurring pattern observed in large-scale multilingual AI deployments: systems that appear to perform well according to aggregate metrics can simultaneously provide significantly degraded experiences for specific user populations. These failures are often not visible in standard evaluation frameworks and therefore persist unaddressed over time.

The concept of Feedback-Induced Distribution Shift (FIDS) is introduced to formalize this phenomenon. The three subtypes, Aggregate Masking Effect (AME), Low-Resource Locale Collapse (LRLC), and Aspirational Deferral Lock (ADL), capture distinct but interacting mechanisms through which performance disparities emerge, compound, and remain unresolved in practice, and each is now backed by a proposition with a short, direct proof rather than an illustrative claim alone. Together, they provide a structured lens for understanding how system-level, feedback-driven dynamics can produce uneven outcomes across linguistic populations.

RL-Stable Deployment (RLSD) is proposed as a complementary mitigation framework that operates at inference time to detect and respond to these dynamics. While not a substitute for improved organizational practices or evaluation standards, it offers a practical mechanism for reducing the impact of FIDS in deployed systems.

More broadly, this work highlights a fundamental limitation in how multilingual AI systems are currently evaluated. Metrics that emphasize aggregate performance can obscure meaningful disparities in user experience, particularly in low-resource or underrepresented contexts. As a result, systems may be considered successful despite systematically underperforming for segments of the populations they serve.

Addressing this challenge requires a shift in both measurement and system design. Evaluation frameworks must move beyond aggregate summaries to account for performance across user groups, including worst-case outcomes. Deployment architectures must incorporate mechanisms for detecting and correcting disparities as they emerge. Ultimately, improving multilingual AI quality depends not only on better models, but on better alignment between what systems optimize and what users actually experience.

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